Beacon Network v2 - Cutting the gemstones

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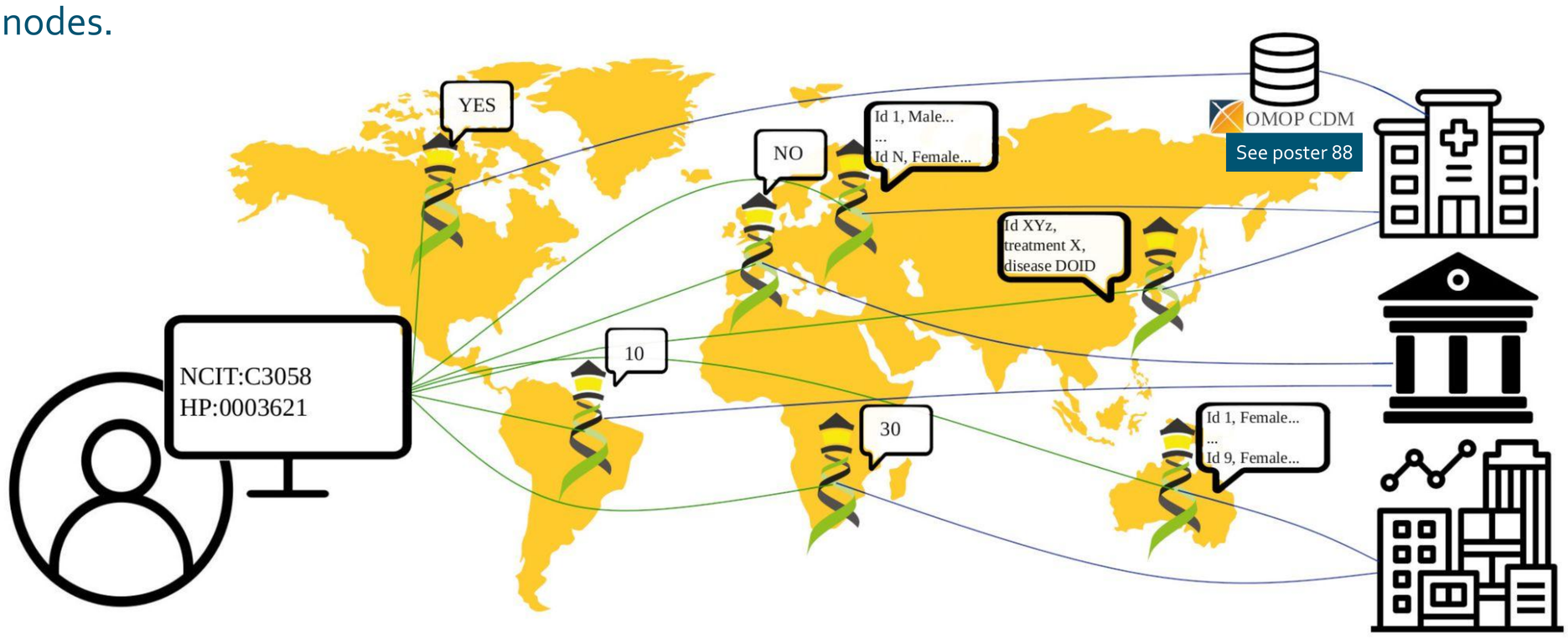
The ELIXIR Beacon Network (EBN) v2 is a **federated data discovery** solution that comprehends Beacon v2 protocol providing an **aggregation of results** obtained from Beacons, both v1 and v2, participating in the network. The current implementation of the EBN is jointly developed by the Barcelona Supercomputing Center (BSC) and the Centre for Genomic Regulation (CRG). The BN aggregator serves to programmatically query the subscribed beacons and gather all answers while the revamped website offers a visual interface to carry out the different queries.

Here we present the first full **prototype** of the ELIXIR Beacon Network. This implementation will help to develop a network of Beacons in a controlled fashion, with close interaction between the network infrastructure developers and the participating nodes.

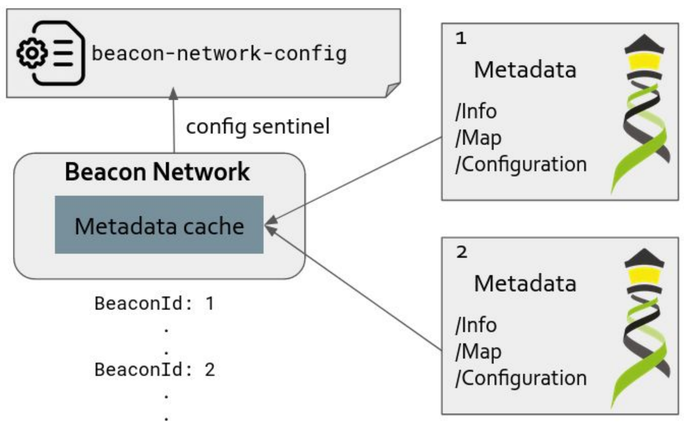


Registry

Check availability of the Beacons and gather their basic information

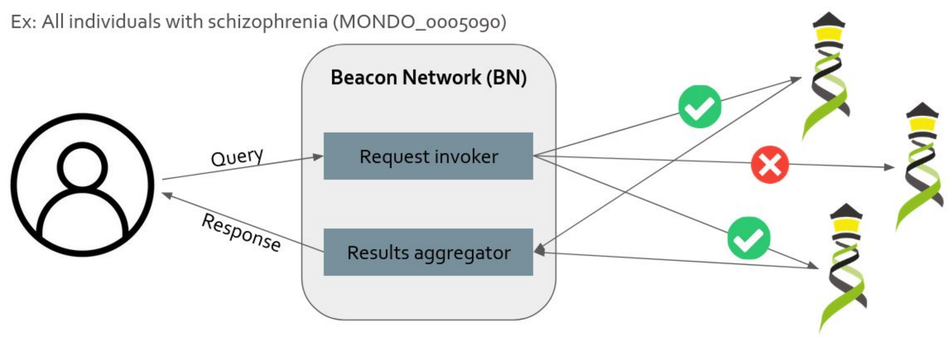
Collect metadata:

- Entry types
- Schemas
- Filtering terms



Aggregator

Query Beacons and gather their results

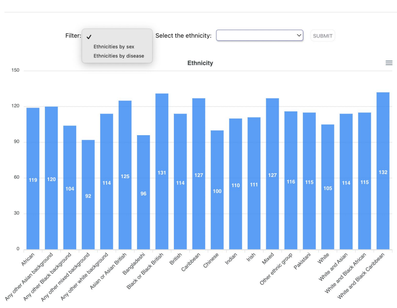


The aggregator has a **query dispatching** policy that checks the availability of entry types, filtering terms, and supported schemas to know what instances can be queried. After querying the Beacons independently, the results are aggregated.

Demo:

Backend: beacons.bsc.es/beacon-network/v2.0.0/

FrontEnd: beacon-network-demo.ega-archive.org/



Cohorts example

Adding new Beacons

Before putting any new Beacon in the network, they should pass through the **Specification Verifier.**

This would make sure that the BN won't find any problem when adding this new instance to the network. To put a Beacon in the ELIXIR Beacon Network you just need to write the URL in a configuration file. Then, the new Beacon will be added automatically.

Check the documentation for more information.

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